[1] The company IBM Japan and four other partners are developing a smart suitcase that uses Artificial Intelligence (Al) to help people with visual disabilities to travel alone with greater freedom and security.

[2] Visually-impaired people usually face problems using public transport and accessing other services. Luckily, there are canes and guide dogs to lead them around obstacles. **These aids** will soon be complemented with smart suitcases which would help them to travel independently.

[3] Chicko Asakawa, an IBM Japan fellow who has vision problems, walks with the assistance of the first prototype of a robotic suitcase equipped with Al. The smart suitcase, which has been inspired by a proposal from Chieko, is equipped with a sensor, a camera, and a battery to detect obstacles and alert its users through tactile sensation. According to its designers, it will also notify users if friends approach or if there are cafés and other facilities nearby through an audio system.

[4] In an interview with a Japanese newspaper,Chieko said that she got the idea when she was pushing a suitcase in front of her during a business trip. She thought that visually impaired people could walk safely if a suitcase would scan the surroundings for danger.

[5] Different companies are contributing to the development of the suitcase. IBM Japan will be responsible for Artificial Intelligence integrated into the suitcase. For its part, Alps Alpine will be responsible for tactile technology, Omron for image recognition and detection techniques, Shimizu for the navigation system, and Mitsubishi Motors for automotive technologies.

[6] The smart suitcase will be used at airports, shopping centers, and in other indoor areas. It will later be improved so that users can use it in outdoor areas. **The device** is expected to be on the market in 2022.

I. COMPREHENSION (15 points)

NB. Base all your answers on the text

A. CHOOSE THE BEST ANSWER: (1 pt)

The text is about:

a. an invention for the visually-impaired.

b. a visually-impaired Japanese woman.

c. a famous robotics company in Japan.

The best answer is: .....

B. ANSWER THESE QUESTIONS: (3 pts)

1. When did Chieko first have the idea of the smart suitcase?

.....

……

2. What will be Alps Alpine's contribution to the design of the smart suitcase?

.....

.....

3. When will people be able to buy the smart suitcase?

.....

.....

C. FILL IN THE CHART WITH THE RIGHT INFORMATION FROM THE TEXT. (1 pt)

Name of company	**Contribution to design the smart suitcase**
1. IBM Japan	……
2. Shimizu	……
3. Omron	……

D. ARE THESE STATEMENTS TRUE OR FALSE? JUSTIFY (3 pts)

1. Guide dogs help visually-impaired people to move easily

.....

.....

2. The smart suitcase will be able to recognize people coming towards its users.

……

……

3. Mitsubishi Motors technology will allow the smart suitcase to recognize images.

.....

.....

E. COMPLETE THE FOLLOWING SENTENCES WITH INFORMATION FROM THE TEXT: (2 pts)

1. Having access to public services is ……

……

2. The ……..... will allow the smart suitcase to notify its users.

F. FIND IN THE TEXT WORDS THAT MEAN ALMOST THE SAME AS: (3 pts)

1. without any help (paragraph 2) : ……

2. model (paragraph 3) : ……

3. Risk (paragraph 4): ……

G. WHAT DO THE UNDERLINED WORDS IN THE TEXT REFER TO? (2 pts)

1. These aids: (paragraph 2): .....

2. The device: (paragraph 6): .....

II. LANGUAGE (15 points)

A. FILL IN THE GAPS WITH THE CORRECT WORDS FROM THE LISTS: (2 pts)

1. Brenda missed her class ..... she was taking part in a sports competition

therefore - because - whereas

2. The vaccine ..... Our laboratory has developed is effective.

whose - whom - which

B. GIVE THE CORRECT FORM OF THE WORDS IN BRACKETS: (2 pts)

1. Their business failed because of poor (manage) ..... and financial problems.

2. Lord of the Rings is one of the most (success) ..... movies I have ever seen.

C. FILL IN THE BLANKS WITH THE CORRECT WORDS FROM THE LISTS: (2 pts)

1. All the guests appreciated Tom's sense of ..... except for Caren.

joke - humor - amusement

2. Scientists will ..... conducting research till they find a cure for cancer.

keep on - log on - try on

D. REWRITE THE SENTENCES AS INDICATED: (3 pts)

1. The jury has interviewed all the candidates online.

All the candidates ……

2. "If I were you, I would buy a laptop instead of a tablet," Danny said.

Danny advised his friend …….

3. Although Saida is from a poor background, she managed to pursue her higher education.

In spite of .....

E. PUT THE VERBS IN BRACKETS IN THE CORRECT TENSE: (2 pts)

1. By the end of this year, Mrs. Simpson (serve) …….……. in the local council for 20 years

2. While Anna (work) ……..……. on her language project, her brother was playing games.

F. MATCH EACH EXPRESSION TO ITS APPROPRIATE FUNCTION: (2 pts)

Expressions	Functions
1. "Why don't you try another job? " 2. "I don't share the same view with you" 3. "Can you give me more details about the topic?' 4. "What do you think of online learning?"	a. asking for opinion b. making a request c. disagreeing d. making a suggestion e. complaining

1. ……. 2. …… 3. …….. 4. …….

G.COMPLETE THE FOLLOWING EXCHANGES: (2 pts)

1. Manager: I'm afraid the report you sent me is full of mistakes.

Pat: .....

2. Sandy: My computer can't read PDF documents. What should I do?

Casey: .....

DO TASKS 1 AND 2

TASK 1: Complete the following paragraph (4 pts)

Many tourists like to visit Morocco for different reasons. First, ...

. . . . . .

. . . . . .

. . . . . .

. . . . . .

. . . . . .

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TASK 2: (6 pts)

Modern technology has facilitated our lives in many ways.

Write a short article for your school magazine about how modern technology has made your life easier.

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KEY AND MARKING SCALE

Accept any appropriate answer not mentioned in this key.

I. COMPREHENSION (15 POINTS)

Testees should not be penalised for spelling or grammar mistakes in the comprehension section.

BEST ANSWER: (1 pt) a. an invention for the visually-impaired.

WH QUESTIONS: (3 pts)

1. When she was pushing a suitcase in front of her during a business trip.
2. It will be responsible for the tactile technology.
3. In 2022.

CHART FILLING: (1 pt)

IBM Japan : the Artificial Intelligence integrated into the suitcase.
Shimizu : the navigation system.
Omron : image recognition and detection techniques.

TRUE / FALSE: (3 pts)

1. **True.** "There are canes and guide dogs to lead them around obstacles." (paragraph 2)
2. **True.** "It will also notify users if friends approach." (paragraph 3)
3. **False.** Mitsubishi Motors is responsible for automotive technologies; *Omron* is responsible for image recognition. (paragraph 5)

SENTENCE COMPLETION: (2 pts) 1. ...a problem for visually-impaired people. 2. audio system

RIGHT WORD MEANING: (3 pts) 1. independently 2. prototype 3. danger

WORD REFERENCE: (2 pts) 1. (The) canes and (the) guide dogs 2. The smart suitcase

II. LANGUAGE (15 POINTS)

GAP FILLING: (2 pts) 1. because 2. which

WORD FORMATION: (2 pts) management · successful

GAP FILLING: (2 pts) 1. humor 2. keep on

REWRITING SENTENCES: (3 pts)

1. All the candidates **have been interviewed online by the jury**.
2. Danny advised his friend **to buy a laptop instead of a tablet**.
3. In spite of **being from a poor background, Saida managed to pursue her higher education**.

VERB TENSES: (2 pts) will have served · was working

MATCHING: (2 pts) 1. d. making a suggestion 2. c. disagreeing 3. b. making a request 4. a. asking for opinion

EXCHANGES: (2 pts)

1. I'm terribly sorry. I'll check it again and send you a corrected version.
2. You should install a PDF reader. / Why don't you download a free PDF reader?

III. WRITING (10 POINTS; task 1: 4 pts; task 2: 6 pts)

The following criteria should be respected while scoring the writing tasks.

Scoring criteria	Task 1	Task 2
Relevance of content to the task	2 pts	3 pts
Appropriateness and variety of vocabulary	1 pt	2 pts
Grammar and mechanics	1 pt	1 pt

Model answers (any appropriate answer is accepted)

Task 1: Many tourists like to visit Morocco for different reasons. First, the country has a beautiful and varied nature, with beaches, mountains and the desert. Second, its historic cities, such as Marrakech and Fez, attract visitors who love culture and architecture. Moreover, Moroccan food is delicious and famous all over the world. Finally, Moroccan people are very friendly and welcoming. For all these reasons, Morocco is an unforgettable destination.

Task 2: How Technology Has Made My Life Easier Modern technology has changed our lives in many positive ways. In my own life, it has made many things easier and faster. First of all, the Internet helps me with my studies. I can search for information in seconds and watch online lessons when I do not understand something. Moreover, my smartphone allows me to keep in touch with my family and friends wherever they are. In addition, I can use apps to translate words, find directions and even learn languages. Technology also saves time: I do not need to go to the bank or to the shops for many things anymore, because I can pay and order online. In conclusion, modern technology has made my life easier, but we must use it wisely and not spend too much time on screens.